SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: B. Tech. Stream: Data Science Year: II Semester: IV

Subject: Stochastic Processes and Applications Time: 3 hrs (10:00am to 1:00pm)

Date: 17 / 04 / 2024 No. of Pages: 09

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1; SO-1; BL-2	a.	Random variable x has CDF $$F_x(x) = \begin{cases} 0 & x < 0, \\ x/2 & 0 \le x \le 2, \\ 1 & x > 2, \end{cases}$$ 1 - What is E[X]? 2 - What is Var[X]?	[5]
CO-2; SO-1; BL-2	b.	The number of airplanes J that arrive at the airport in time t minutes is a Poisson random variable with E[J] = t. Find t such that P[J > 0] = 0.8.	[5]
CO-4; SO-1; BL-3	c.	Sketch the transition diagrams of the following Markov chain. $$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$	[5]
CO-3; SO-1; BL-3	d.	Several c cashiers at a supermarket can be modeled as a c server queue with infinite capacity. Assuming the service times are independent exponential random variables with mean μ seconds. a) What type of queue system does this problem represent?	[5]

		b) Sketch a Markov chain for the system.	
		Answer any four from Q2 to Q7	
Q2 CO-3; SO-1; BL-3	a	Random variables X and Y have joint PDF $f_{X,Y}(x,y) = \begin{cases} cx & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & otherwise \end{cases}$ Find the constant c.	[7]
CO-3; SO-1; BL-2	b	Referring **Q.2.a** Find the marginal PDF $f_X(x)$.	[7]
CO-3; SO-1; BL-2	c	Referring **Q.2.a** Are X and Y independent? Justify your answer.	[6]
Q3 CO-2; SO-1; BL-3	a	Queries presented to a computer database are a Poisson process of rate λ= 6 queries per minute. An experiment consists of monitoring the database for m minutes and recording N(m), the number of queries presented. The answer to each of the following questions can be expressed in terms of the PMF PN(m)(k) = P[N(m) = k]. (a) What is the probability of no queries in a one-minute interval? (b) What is the probability of exactly six queries arriving in a one-minute interval? (c) What is the probability of exactly three queries arriving in a one-half-minute interval?	[10]
CO-4; SO-1; BL-3	b	A continuous function of a random variable X has the p.d.f $f(x) = kx^2e^{-x}$, x>=0. Calculate k, mean and variance.	[10]
Q4 CO-3; SO-1; BL-3	a	Determine whether each of these statements is true or false. Justify your answer for full credit. (a) If X(t) and Y (t) are independent wide sense stationary processes, then the process W(t) = X(t)Y(t). Is a wide-sense stationary: (b) Given a Gaussian process X(t), then 2X(t) − 3X(t − 1) is not a Gaussian process.	[10]
CO-3; SO-1; BL-3	b	A random process is given by $X(t) = A\cos(\omega t) + B\sin(\omega t)$ where A and B are random variables. Show that $X(t)$ is wide sense stationary if the expectations are given (i) $E(A) = E(B) = 0$ (ii) $E(A^2) = E(B^2)$ and (iii) $E(AB) = 0$.	[10]
Q5 CO-3; SO-1; BL-3	a	Consider a random walk on the finite states {-2, -1, 0,1,2}. If the process is in state i (i=-1,0,1) at time n, then it moves to either i-1 or i+1 at time n+1 with equal probability. If the process is in state -2 or 2 at time n, then it moves to state -1, 0, or 1 at time n+1 with equal probability. 1- Write the transition probability matrix P for this random walk.	[20]

		2- Compute the stationary probability $\pi = (\pi_{-2}, \pi_{-1}, \pi_0, \pi_1, \pi_2)$ of this random walk. [Hint: You can use the symmetry so that $\pi_{-i} = \pi_i$ for i=1,2.]	
Q6 CO-3; SO-1; BL-3	a	Consider the discrete time Markov chain with state space Ω={0,1,2,3,4} with the one-step transition probability matrix $$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.75 & 0.25 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$ a-Determine the period of state 1. b-Determine the period of state 2. c-Is the chain irreducible? Justify your answer	[10]
CO-3; SO-1; BL-3	b	The transition probability matrix (TPM) of Marcov chain $\{X_n\}_{n=1,2,3,..}$ with three states is as follows. $$\begin{array}{cc} & \textbf{States of } X_{n+1} \\ & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \textbf{States of } X_n \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \end{array}$$ The initial distribution P^0 is as follows. $$\begin{array}{cc} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ P^0 = & \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix} \end{array}$$ Find $P(X_2 = 3)$, $P(X_2 = 2)$, and $P(X_3 = 2,\ X_2 = 3,\ X_1 = 3,\ X_0 = 2)$	[10]
Q7 CO-4; SO-1; BL-2	a	Compare Random Process and Random Variable. Explain the Classification of Random Processes	[10]
CO-2; SO-1; BL-3	b	The average number of visits at an amusement park is 30 children in one hour. What is the probability that (i) there are no visits in a 3-minutes time span, (ii) at least 5 visits in a 5-minutes time span?	[10]

Reference Notes:

X is a Poisson (α) random variable if the PMF of X has the form

$$P_X(x) = \begin{cases} \alpha^x e^{-\alpha}/x! & x = 0, 1, 2, \ldots, \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter α is in the range $\alpha > 0$.

- *In this time interval,* ***the number of arrivals X has a Poisson PMF with*** ***$\alpha = \lambda T$***

The expected value of X is

$$E[X] = \mu_X = \sum_{x \in S_X} x P_X(x).$$

The expected value of a continuous random variable X is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

For any random variable X,

(a) $E[X - \mu_X] = 0$,

(b) $E[aX + b] = a\,E[X] + b$,

(c) $\mathrm{Var}[X] = E[X^2] - \mu_X^2$,

(d) $\mathrm{Var}[aX + b] = a^2\,\mathrm{Var}[X]$.

X is a uniform (a,b) random variable if the PDF of X is

$$f_X(x) = \begin{cases} 1/(b-a) & a \le x < b, \\ 0 & \text{otherwise,} \end{cases}$$

where the two parameters are $b > a$.

If X is a uniform (a,b) random variable,

- The CDF of X is $$F_X(x) = \begin{cases} 0 & x \le a, \\ (x-a)/(b-a) & a < x \le b, \\ 1 & x > b. \end{cases}$$
- The expected value of X is $\quad \mathrm{E}[X] = (b+a)/2.$
- The variance of X is $\quad \mathrm{Var}[X] = (b-a)^2/12.$

X *is an* exponential (λ) random variable *if the PDF of X is*

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \ge 0, \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter $\lambda > 0$.

X is a Gaussian (μ,σ) random variable if the PDF of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$

where the parameter μ can be any real number and the parameter $\sigma > 0$.

If X is Gaussian (μ,σ), $Y = aX + b$ is Gaussian $(a\mu + b, a\sigma)$.

$X(t)$ *is a* wide sense stationary stochastic process *if and only if for all* t,

$$\mathsf{E}[X(t)] = \mu_X, \quad \text{and} \quad R_X(t,\tau) = R_X(0,\tau) = R_X(\tau).$$

....

X_n *is a* wide sense stationary random sequence *if and only if for all* n,

$$\mathsf{E}[X_n] = \mu_X, \quad \text{and} \quad R_X[n,k] = R_X[0,k] = R_X[k].$$

The cross-correlation *of continuous-time random processes* $X(t)$ *and* $Y(t)$ *is*

$$R_{XY}(t,\tau) = \mathsf{E}[X(t)Y(t+\tau)].$$

....

The cross-correlation *of random sequences* X_n *and* Y_n *is*

$$R_{XY}[m,k] = \mathsf{E}\left[X_m Y_{m+k}\right].$$

For a finite Markov chain, the n-step transition probabilities satisfy

$$P_{ij}(n+m) = \sum_{k=0}^{K} P_{ik}(n)P_{kj}(m), \qquad \mathbf{P}(n+m) = \mathbf{P}(n)\mathbf{P}(m).$$

For a finite Markov chain with transition matrix $\mathbf{P}$, *the n-step transition matrix is*

$$\mathbf{P}(n) = \mathbf{P}^n.$$

The state probabilities $p_j(n)$ at time n can be found by either one iteration with the n-step transition probabilities:

$$p_j(n) = \sum_{i=0}^{K} p_i(0)P_{ij}(n), \qquad \mathbf{p}'(n) = \mathbf{p}'(0)\mathbf{P}^n,$$

or n iterations with the one-step transition probabilities:

$$p_j(n) = \sum_{i=0}^{K} p_i(n-1)P_{ij}, \qquad \mathbf{p}'(n) = \mathbf{p}'(n-1)\mathbf{P}.$$

If a finite Markov chain with transition matrix $\mathbf{P}$ *and initial state probability* $\mathbf{p}(0)$ *has limiting state probability vector* $\boldsymbol{\pi} = \lim_{n\to\infty} \mathbf{p}(n)$, *then*

$$\boldsymbol{\pi}' = \boldsymbol{\pi}'\mathbf{P}.$$

For an irreducible, aperiodic, finite Markov chain with states $\{0, 1, \ldots, K\}$, *the limiting* n*-step transition matrix is*

$$\lim_{n\to\infty} \mathbf{P}^n = \mathbf{1}\boldsymbol{\pi}' = \begin{bmatrix} \pi_0 & \pi_1 & \cdots & \pi_K \\ \pi_0 & \pi_1 & \cdots & \pi_K \\ \vdots & & \ddots & \vdots \\ \pi_0 & \pi_1 & & \pi_K \end{bmatrix}$$

where $\mathbf{1}$ *is the column vector* $\begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}'$ *and* $\boldsymbol{\pi} = \begin{bmatrix} \pi_0 & \cdots & \pi_K \end{bmatrix}'$ *is the unique vector satisfying*

$$\boldsymbol{\pi}' = \boldsymbol{\pi}'\mathbf{P}, \qquad \boldsymbol{\pi}'\mathbf{1} = 1.$$

Consider an irreducible, aperiodic, finite Markov chain with transition probabilities $\{P_{ij}\}$ *and stationary probabilities* $\{\pi_i\}$. *For any partition of the state space into mutually exclusive subsets* S *and* S',

$$\sum_{i\in S}\sum_{j\in S'} \pi_i P_{ij} = \sum_{j\in S'}\sum_{i\in S} \pi_j P_{ji}.$$

For an irreducible, recurrent, periodic, finite Markov chain with transition probability matrix $\mathbf{P}$, *the stationary probability vector* $\boldsymbol{\pi}$ *is the unique non-negative solution of*

$$\boldsymbol{\pi}' = \boldsymbol{\pi}'\mathbf{P}, \qquad \boldsymbol{\pi}'\mathbf{1} = 1.$$

For a Markov chain with recurrent communicating classes $C_1, \ldots, C_m$, *let* $\boldsymbol{\pi}^{(k)}$ *denote the limiting state probabilities associated with class* C_k. *Given that the system starts in a transient state* i, *the limiting probability of state* j *is*

$$\lim_{n\to\infty} P_{ij}(n) = \pi_j^{(1)}\,\mathrm{P}\left[B_{i1}\right] + \cdots + \pi_j^{(m)}\,\mathrm{P}\left[B_{im}\right]$$

where $\mathrm{P}[B_{ik}]$ *is the conditional probability that the system enters class* C_k.

For an irreducible, positive recurrent continuous-time Markov chain, the state probabilities satisfy

$$\lim_{t\to\infty} p_j(t) = p_j, \qquad \text{or, in vector form,} \quad \lim_{t\to\infty} \mathbf{p}(t) = \mathbf{p}$$

where the limiting state probabilities are the unique solution to

$$\sum_i r_{ij} p_i = 0, \qquad \text{or, in vector form,} \quad \mathbf{p}'\mathbf{R} = \mathbf{0}',$$

$$\sum_j p_j = 1, \qquad \text{or, in vector form,} \quad \mathbf{p}'\mathbf{1} = 1.$$

For a birth-death queue with arrival rates λ_i and service rates μ_i, the stationary probabilities p_i satisfy

$$p_{i-1}\lambda_{i-1} = p_i \mu_i, \qquad \sum_{i=0}^{\infty} p_i = 1.$$

The M/M/1 queue with arrival rate $\lambda > 0$ and service rate μ, $\mu > \lambda$, has limiting state probabilities

$$p_n = (1-\rho)\rho^n, \qquad n = 0, 1, 2, \ldots$$

where $\rho = \lambda/\mu$.

The M/M/∞ queue with arrival rate $\lambda > 0$ and service rate $\mu > 0$ has limiting state probabilities

$$p_n = \begin{cases} \rho^n e^{-\rho}/n! & n = 0, 1, 2, \ldots, \\ 0 & \text{otherwise,} \end{cases}$$

where $\rho = \lambda/\mu$.

For the M/M/c/c queue with arrival rate λ and service rate μ, the limiting state probabilities satisfy

$$p_n = \begin{cases} \dfrac{\rho^n/n!}{\sum_{j=0}^{c} \rho^j/j!} & j = 0, 1, \ldots, c, \\ 0 & \textit{otherwise}, \end{cases}$$

where $\rho = \lambda/\mu$.

For the M/M/1 queue, the stationary probability of state n is $p_n = (1-\rho)\rho^n$. Hence the PMF of N is

$$P_N(n) = \begin{cases} (1-\rho)\rho^n & n = 0, 1, 2, \ldots, \\ 0 & \text{otherwise.} \end{cases} \tag{93}$$

The expected number in the queue is

$$\mathrm{E}[N] = \sum_{n=0}^{\infty} n(1-\rho)\rho^n = \rho \sum_{n=1}^{\infty} n(1-\rho)\rho^{n-1} = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu-\lambda}. \tag{94}$$